

# Practice from Assignment 1 and 2

Thursday, May 8, 2025

11:03 AM

[Assg 1](#) problems: 2,4,6,10,12,16,18,20,22,24 -- will be evaluated in quiz 1

## PROBLEM SET 9.4

### 1–8 SCALAR FIELDS IN THE PLANE

Let the temperature  $T$  in a body be independent of  $z$  so that it is given by a scalar function  $T = T(x, y)$ . Identify the isotherms  $T(x, y) = \text{const.}$  Sketch some of them.

1.  $T = x^2 - y^2$
2.  $T = xy$
3.  $T = 3x - 4y$
4.  $T = \arctan(y/x)$
5.  $T = y/(x^2 + y^2)$
6.  $T = x/(x^2 + y^2)$
7.  $T = 9x^2 + 4y^2$

8. **CAS PROJECT. Scalar Fields in the Plane.** Sketch or graph isotherms of the following fields and describe what they look like.

- |                           |                      |
|---------------------------|----------------------|
| (a) $x^2 - 4x - y^2$      | (b) $x^2y - y^3/3$   |
| (c) $\cos x \sinh y$      | (d) $\sin x \sinh y$ |
| (e) $e^x \sin y$          | (f) $e^{2x} \cos 2y$ |
| (g) $x^4 - 6x^2y^2 + y^4$ | (h) $x^2 - 2x - y^2$ |

### 9–14 SCALAR FIELDS IN SPACE

What kind of surfaces are the level surfaces  $f(x, y, z) = \text{const.}$ ?

- |                           |                                |
|---------------------------|--------------------------------|
| 9. $f = 4x - 3y + 2z$     | 10. $f = 9(x^2 + y^2) + z^2$   |
| 11. $f = 5x^2 + 2y^2$     | 12. $f = z - \sqrt{x^2 + y^2}$ |
| 13. $f = z - (x^2 + y^2)$ | 14. $f = x - y^2$              |

### 15–20 VECTOR FIELDS

Sketch figures similar to Fig. 198. Try to interpret the field of  $\mathbf{v}$  as a velocity field.

15.  $\mathbf{v} = \mathbf{i} + \mathbf{j}$
16.  $\mathbf{v} = -y\mathbf{i} + x\mathbf{j}$

### 22–25 DIFFERENTIATION

22. Find the first and second derivatives of  $\mathbf{r} = [3 \cos 2t, 3 \sin 2t, 4t]$ .
23. Prove (11)–(13). Give two typical examples for each

Sketch figures similar to Fig. 19.07 (1) to interpret the flow of  $\mathbf{v}$  as a velocity field.

15.  $\mathbf{v} = \mathbf{i} + \mathbf{j}$

17.  $\mathbf{v} = x\mathbf{j}$

19.  $\mathbf{v} = x\mathbf{i} - y\mathbf{j}$

21. **CAS PROJECT. Vector Fields.** Plot by arrows:

(a)  $\mathbf{v} = [x, x^2]$

(b)  $\mathbf{v} = [1/y, 1/x]$

(c)  $\mathbf{v} = [\cos x, \sin x]$

(d)  $\mathbf{v} = e^{-(x^2+y^2)}[x, -y]$

16.  $\mathbf{v} = -y\mathbf{i} + x\mathbf{j}$

18.  $\mathbf{v} = x\mathbf{i} + y\mathbf{j}$

20.  $\mathbf{v} = y\mathbf{i} - x\mathbf{j}$

22. Find the first and second derivatives of  $\mathbf{r} = [3 \cos 2t, 3 \sin 2t, 4t]$ .

23. Prove (11)–(13). Give two typical examples for each formula.

24. Find the first partial derivatives of  $\mathbf{v}_1 = [e^x \cos y, e^x \sin y]$  and  $\mathbf{v}_2 = [\cos x \cosh y, -\sin x \sinh y]$ .

25. **WRITING PROJECT. Differentiation of Vector Functions.** Summarize the essential ideas and facts and give examples of your own.

[Assg 2](#) problems: 2,4,12,14, 24,31,35 --will be evaluated in quiz 1.

### 1–10 PARAMETRIC REPRESENTATIONS

What curves are represented by the following? Sketch them.

1.  $[3 + 2 \cos t, 2 \sin t, 0]$

2.  $[a + t, b + 3t, c - 5t]$

3.  $[0, t, t^3]$

4.  $[-2, 2 + 5 \cos t, -1 + 5 \sin t]$

5.  $[2 + 4 \cos t, 1 + \sin t, 0]$

6.  $[a + 3 \cos \pi t, b - 2 \sin \pi t, 0]$

7.  $[4 \cos t, 4 \sin t, 3t]$

8.  $[\cosh t, \sinh t, 2]$

9.  $[\cos t, \sin 2t, 0]$

10.  $[t, 2, 1/t]$

20. Intersection of  $2x - y + 3z = 2$  and  $x + 2y - z = 3$ .

21. **Orientation.** Explain why setting  $t = -t^*$  reverses the orientation of  $[a \cos t, a \sin t, 0]$ .

22. **CAS PROJECT. Curves.** Graph the following more complicated curves:

(a)  $\mathbf{r}(t) = [2 \cos t + \cos 2t, 2 \sin t - \sin 2t]$  (Steiner's hypocycloid).

(b)  $\mathbf{r}(t) = [\cos t + k \cos 2t, \sin t - k \sin 2t]$  with  $k = 10, 2, 1, \frac{1}{2}, 0, -\frac{1}{2}, -1$ .

(c)  $\mathbf{r}(t) = [\cos t, \sin 5t]$  (a Lissajous curve).

(d)  $\mathbf{r}(t) = [\cos t, \sin kt]$ . For what  $k$ 's will it be closed?

(e)  $\mathbf{r}(t) = [R \sin \omega t + \omega R t, R \cos \omega t + R]$  (cycloid).

### 11–20 FIND A PARAMETRIC REPRESENTATION

11. Circle in the plane  $z = 1$  with center  $(3, 2)$  and passing through the origin.

12. Circle in the  $yz$ -plane with center  $(4, 0)$  and passing through  $(0, 3)$ . Sketch it.

13. Straight line through  $(2, 1, 3)$  in the direction of  $\mathbf{i} + 2\mathbf{j}$ .

14. Straight line through  $(1, 1, 1)$  and  $(4, 0, 2)$ . Sketch it.

15. Straight line  $y = 4x - 1, z = 5x$ .

16. The intersection of the circular cylinder of radius 1 about the  $z$ -axis and the plane  $z = y$ .

17. Circle  $\frac{1}{2}x^2 + y^2 = 1, z = y$ .

18. Helix  $x^2 + y^2 = 25, z = 2 \arctan(y/x)$ .

19. Hyperbola  $4x^2 - 3y^2 = 4, z = -2$ .

### 24–28 TANGENT

Given a curve  $C: \mathbf{r}(t)$ , find a tangent vector  $\mathbf{r}'(t)$ , a unit tangent vector  $\mathbf{u}'(t)$ , and the tangent of  $C$  at  $P$ . Sketch curve and tangent.

24.  $\mathbf{r}(t) = [t, \frac{1}{2}t^2, 1]$ ,  $P: (2, 2, 1)$

25.  $\mathbf{r}(t) = [10 \cos t, 1, 10 \sin t]$ ,  $P: (6, 1, 8)$

26.  $\mathbf{r}(t) = [\cos t, \sin t, 9t]$ ,  $P: (1, 0, 18\pi)$

27.  $\mathbf{r}(t) = [t, 1/t, 0]$ ,  $P: (2, \frac{1}{2}, 0)$

28.  $\mathbf{r}(t) = [t, t^2, t^3]$ ,  $P: (1, 1, 1)$

### 29–32 LENGTH

Find the length and sketch the curve.

29. Catenary  $\mathbf{r}(t) = [t, \cosh t]$  from  $t = 0$  to  $t = 1$ .

30. Circular helix  $\mathbf{r}(t) = [4 \cos t, 4 \sin t, 5t]$  from  $(4, 0, 0)$  to  $(4, 0, 10\pi)$ .

31. **Circle**  $\mathbf{r}(t) = [a \cos t, a \sin t]$  from  $(a, 0)$  to  $(0, a)$ .
32. **Hypocycloid**  $\mathbf{r}(t) = [a \cos^3 t, a \sin^3 t]$ , total length.
33. **Plane curve.** Show that Eq. (10) implies  $\ell = \int_a^b \sqrt{1 + y'^2} dx$  for the length of a plane curve  $C: y = f(x), z = 0$ , and  $a = x = b$ .
34. **Polar coordinates**  $\rho = \sqrt{x^2 + y^2}, \theta = \arctan(y/x)$  give

$$\ell = \int_{\alpha}^{\beta} \sqrt{\rho^2 + \rho'^2} d\theta,$$

where  $\rho' = d\rho/d\theta$ . Derive this. Use it to find the total length of the **cardioid**  $\rho = a(1 - \cos \theta)$ . Sketch this curve. *Hint.* Use (10) in App. 3.1.

### 35–46 CURVES IN MECHANICS

Forces acting on moving objects (cars, airplanes, ships, etc.) require the engineer to know corresponding *tangential* and *normal accelerations*. In Probs. 35–38 find them, along with the *velocity* and *speed*. Sketch the path.

35. **Parabola**  $\mathbf{r}(t) = [t, t^2, 0]$ . Find  $\mathbf{v}$  and  $\mathbf{a}$ .
36. **Straight line**  $\mathbf{r}(t) = [8t, 6t, 0]$ . Find  $\mathbf{v}$  and  $\mathbf{a}$ .
37. **Cycloid**  $\mathbf{r}(t) = (R \sin \omega t + Rt)\mathbf{i} + (R \cos \omega t + R)\mathbf{j}$ .  
This is the path of a point on the rim of a wheel of radius  $R$  that rolls without slipping along the  $x$ -axis.  
Find  $\mathbf{v}$  and  $\mathbf{a}$  at the maximum  $y$ -values of the curve.
38. **Ellipse**  $\mathbf{r} = [\cos t, 2 \sin t, 0]$ .